

set of processes to ensure its usability and functionality for the intended purpose. Behind each of these steps are design, manufacturing and testing procedures. T&M technologies are going to evolve accordingly. There is need for embedded/portable T&M solutions, IoT test solutions with remote troubleshooting capabilities and solutions that can support the latest interfaces and protocols.

It is certain that time-to-market for new devices will continue to tighten, and maintaining high quality will continue to drive additional complexity in these devices. However, all this will be invisible to users. Test techniques and solutions will continue to evolve to ensure that smart device developers and manufacturers can reduce time-to-market and ramp quickly to high-volume manufacturing of quality products.

Connected lifestyles, higher use of data and lower cost of smartphones are some trends in the telecom market. Need for high-speed connectivity will lead to increasing network capacity. Operators are using new technologies like MIMO, VoLTE, LTE-Advanced and LTE Carrier Aggregation to improve spectrum efficiency and customer experience with high data speeds. 4G roll out and 5G-related R&D in telecom create additional opportunities for the T&M industry.

Compared to previous generations of mobile networks, 5G presents many new design and test challenges. While the technologies that will constitute 5G are still being worked on, ITU-R has identified three main usage scenarios for 5G. These are enhanced mobile broadband with up to 20Gbps peak data rate, massive machine-type communications, and ultra-reliable and low latency communications.

To meet the requirements for this next generation of mobile communi-

Market challenges

- Import of cheap and low-quality equipment from China
- Tough price competition
- Low customer budget
- No major incentive/subsidies from government for capital investments in T&M
- High GST rate of 18% levied on T&M industry (appeals have been made to reduce the same to 12%)
- No major incentives for exporters of T&M equipment

SWOT ANALYSIS OF THE INDIAN T&M MARKET

Strength	Weakness	Opportunities	Threats
• Continuous technological advances fuelling the need for testing equipment • Ongoing practice of procuring the most reliable testing equipment • Adoption of best testing practices by end user groups • Stringent quality and safety standards for end products	• Lack of mandatory enforcement of quality standards for end products • Absence of well-documented quality standards for T&M products • No policy boost for capital investment for procuring T&M equipment • Lack of industry awareness about usability and functionality of testing equipment	• Introduction of various government initiatives like Make in India and Digital India resulting in growth of domestic manufacturing in application sectors like defence and automotive • Increasing demand for precision testing throughout the life cycle of electronic products • Proliferation of advanced-level electronic products • Increasing awareness for predictive maintenance among manufacturers	• Various unscrupulous players not adhering to standard engineering and commercial practices • Price-sensitive nature of end user groups • Quality affected because most of the time the government's tender business done only on lowest price (L1) basis

cations, researchers are looking to a wide range of enabling technologies in RF, microwave and millimetre-wave frequencies, ranging from advanced waveforms and multiple access schemes to multiple antenna techniques like massive MIMO and beamforming. Being the first and the best in 5G depends on tools that let you explore new signals, scenarios and topologies.

Automation powered by M2M interaction is the new technology trend in the power sector and, hence, demand. Accordingly, new technological advancements like data logging DMMs and auto power calculating power clamp meters will experience good growth. Online transmission

line monitoring systems, compact Tan-delta testing machines, automatic breaker testing machines and discrete monitoring of power loss at each feeder by discom are some areas of application.

With respect to user experience, touchscreen technologies will revolutionise the T&M environment. A combination of the right hardware with a design for touch user interface (UI) will enable us to use equipment easily and efficiently. Touch UIs are crucial for creating an effective and usable touchscreen environment. It is not enough to simply add a touchscreen to an existing UI. New age T&M equipment needs to be able to identify, isolate and analyse a system under test as quickly and efficiently as possible. Having an effective touchscreen on the instrument can reduce the time it takes to configure the system and to analyse signals. Thus, instruments with better GUIs provide ease of operation and automated testing capabilities, which increase productivity. **EFY**

Major contributors to this report

- Madhukar Tripathi, head - marcom and optical products, Anritsu India Pvt Ltd
- Nasser Jariwala, director - sales and operations (T&M), Rohde & Schwarz India
- V. Vijayakumar, general manager - sales and marketing, Qmax Test Equipments Pvt Ltd
- S.A. Srinivasa Moorthy, group chief technology officer, Precision Group
- Shailendra Kalra, managing director, Livingston India Pvt Ltd